

# The Art of Queueing

Reinforcement Learning for visitor flow at the Gallerie degli Uffizi in Florence

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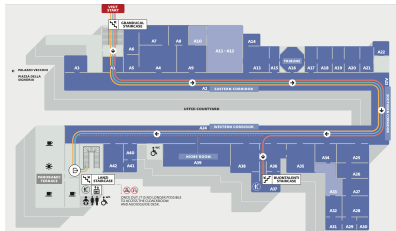
June 2026

# The few rooms everyone queues for

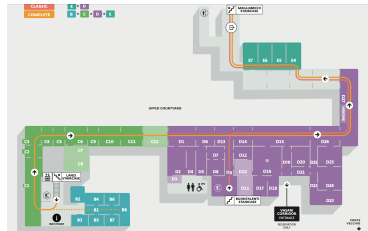
Almost all  $\sim 5,000$  daily visitors come for the same handful of paintings, crammed into three of ninety-eight rooms, so those rooms run far over capacity while the rest sit calm. The crowd is mixed: 60% Instagram tourists, 30% normal, 10% art lovers.



*Second floor (the gated masterpieces)*



*First floor (Caravaggio, free-flow)*



## The problem and the goal

Those three rooms run far over capacity while ninety-five sit half empty: the masterpiece rooms mean long queues and a crush; the rest, deserted galleries. The crowd never coordinates, so every visitor pressing into a packed Botticelli makes it worse for the next, a textbook *congestion externality*.

*Our question: can reinforcement learning do better?* Two goals.

### **The museum.**

Redesign the incentives so the same crowd spreads out, raising welfare without turning anyone away or cutting revenue.

### **The visitor.**

Find the smartest way through the museum despite the crowd: when to book, which masterpieces, how to pace the day.

## Two streams, one project

We split the project into two parts kept separate until the end.

### Stream 1: crowd simulator.

Full 60/30/10 crowd, one day.

- Museum *as-is* versus a Pigovian redesign.
- Adopt only if welfare rises, revenue holds.
- It passes: welfare **+31%**, revenue preserved.

### Stream 2: RL on top (our focus).

Take that museum as fixed; find the *optimal* visitor.

- When to book and which masterpieces.
- How to pace the day.
- All under crowd uncertainty.

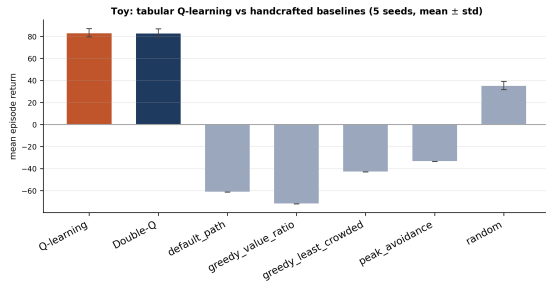


# Foundations: a tabular proof of concept

We prove it on a *12-room toy* with an exact value table. Crowds wave up and down, unannounced.

Tabular *Q-learning* learns *temporal arbitrage*: enter in the lull around the crowd peak.

*Across five seeds*: Q and *Double-Q* tie (83.3 vs 83.1). A deterministic toy has no overestimation bias to remove, exactly as theory predicts.



Learning beats every hand-written rule; the rules even go negative by mistiming the crowd.

## Scaling up: deep RL and action masking

The real museum breaks the table: ninety-eight rooms,  $\sim 100$  moves per step, hundreds of state numbers. We swap it for a neural net (*MaskablePPO*; PPO and DQN as controls).

### Action masking: the rulebook

From any room only a few moves are legal, yet the net has  $\sim 100$  outputs. We set every illegal move to  $-\infty$  before the softmax, so it gets exactly zero probability. No gradient is wasted on moves that do not exist; this state-dependent action set is our single biggest design choice.

$$\pi(a \mid s) = \frac{e^{z_a} \mathbf{1}[a \in \mathcal{A}(s)]}{\sum_b e^{z_b} \mathbf{1}[b \in \mathcal{A}(s)]}.$$

## The reframe: when to book, not where to walk

A visitor has a map and guide, so the route is a shortest path. The real decision is the *booking*, hard for three reasons walking never has:

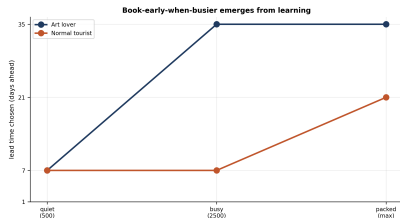
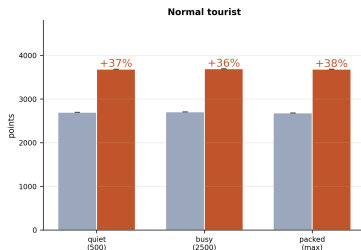
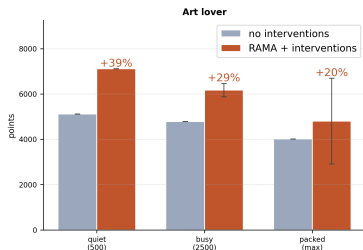
- *Uncertain*: you cannot see how fast the good slots fill.
- *Irreversible*: book late and the masterpiece is gone.
- *Delayed*: you act today; the payoff lands at check-in ( $\gamma = 0.999$ ).

*Baseline (as-is)*: no booking, take masterpieces as found.

*RAMA*: booked slots, pick a lead time of 1, 7, 21 or 35 days.

# Result: book early when busier and it pays

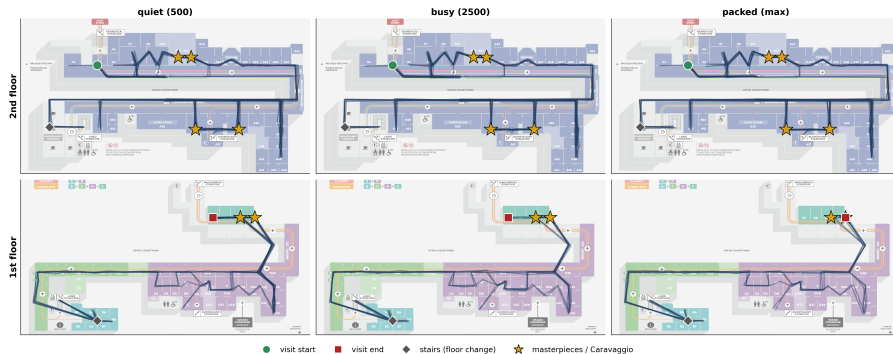
RAMA booking: intervened vs matched baseline (5 seeds, mean  $\pm$  std; 3/3 masterpieces secured)



Both profiles beat the matched no-booking baseline at every crowd level: **+20% to +39%** for the art lover, **+36% to +38%** for the tourist, all three masterpieces secured. The learned lead time *rises with the crowd* (right): book-early-when-busier is not coded in, it emerges from the reward, because booking late at a packed museum misses the rooms.

# What a visit looks like, on the real floor plan

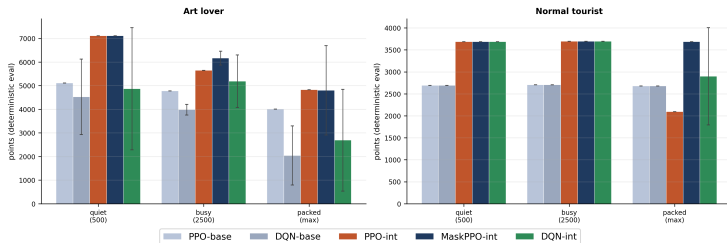
Art lover walking routes (RAMA + interventions) -- one continuous visit, 8 rollouts/panel; It descends floor 2 -> floor 1.



*The art lover's learned visit, eight rollouts per panel, across the three crowd levels. It is one continuous, sensible route: enter at A1, sweep the second-floor masterpieces (gold stars) in their booked windows, descend the staircase to Caravaggio on the first floor and leave before closing. Every step is a one-hop move to a neighbouring room, so there are no teleports.*

# Which algorithm wins and why masking matters

Algorithm comparison: value-based (DQN) vs policy-gradient (PPO / MaskablePPO), baseline vs intervened (5 seeds, mean  $\pm$  std)



- *MaskablePPO* is best or tied in every cell and stable across seeds.
- *Masking is decisive exactly at the hardest cell*: at the packed tourist, unmasked PPO *collapses* to 2093, below the no-booking baseline of 2680, while MaskablePPO holds 3685. At easier crowds the two agree; the gap appears only where precision matters.
- *DQN is crowd-fragile and high-variance* (its baseline falls 4527  $\rightarrow$  2044): taking a max over noisy crowd values overestimates, the very bias the deterministic toy did not have.

## 1: an unlearnable reward

**S** *Symptom.* The agent never left, squatting in one room until closing. Zeroing all reward for not exiting made it *worse*.

**D** *Diagnosis.* An all-or-nothing wipe is *flat* almost everywhere, so the gradient is zero: nothing points to the door.

**F** *Fix.* Use a *dense* egress signal: per-step pressure rising near closing and far from an exit, plus a bonus for leaving.

**L** *Lesson.* Sparse rewards are correct but often unlearnable. Shape a slope the optimiser can climb.

## 2: it could not see what it was deciding

**S** *Symptom.* The rule “stay until satisfied, then move” never stabilised; dwell times wobbled at random.

**D** *Diagnosis.* The right action depended on value already taken from the room, *not in the state*: a hidden variable, a POMDP not the MDP we assumed.

**F** *Fix.* Add a per-room *appreciation-progress* value to the state, so “seen enough?” becomes something the policy can read.

**L** *Lesson.* The state must hold everything the optimal action depends on, or you solve a different problem.



### 3: reward shaping that backfired

**S** *Symptom.* Well-meant terms backfired: a “boredom” penalty caused *looping*, a harsh crowd penalty made a *fake art lover*.

**D** *Diagnosis.* Every term is a new objective to game (reward hacking): looping beat exiting, a crowded Botticelli scored worse than skipping it.

**F** *Fix.* *Minimal* shaping: a *gentle* crowd discount so a crowded masterpiece still beats skipping, no boredom penalty, satiation alone stops lingering.

**L** *Lesson.* Ask of every term: what is the cheapest way to collect it? Is that what you want?

## 4: the do-nothing trap and the book-early trick

**S** *Symptom*. Given a free “decline” button, *both* profiles declined all three masterpieces; book-early stayed only *half* learned.

**D** *Diagnosis*. Declining is safe now, booking pays only later: a do-nothing optimum; its reward lands hundreds of steps away (credit assignment).

**F** *Fix*. Make “always books” a *type trait* with no escape hatch; add an *immediate off-pace penalty* proxying the distant cost.

**L** *Lesson*. A small immediate signal tracking a far-off reward makes the long-horizon lesson learnable: our most decisive trick.

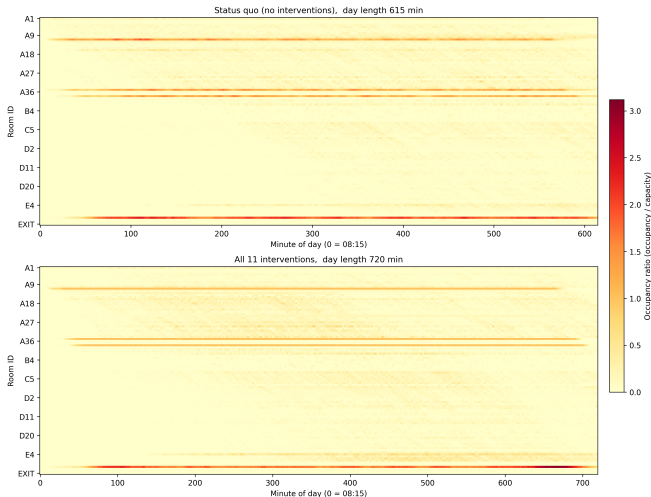
## What five seeds revealed (and why we ran them)

A single training run can be lucky or unlucky, so a single number can mislead. We retrained every cell under *five* training seeds and report the mean and the spread across them.

- *The tourist is rock-stable*: essentially zero variance at every crowd level.
- *The art lover degrades as the museum fills*: the gain falls from +39% to +29% to +20%.
- *The catch*: at *maximum crowd* the art lover is *bimodal*, four of the five seeds reach  $\sim +41\%$  but one **collapses** below baseline, so the cell averages  $+20\%$  with a wide error bar. We report the spread, not the lucky seed.

# The crowd before and after the redesign

Crowd density heatmap: status quo vs all 11 interventions (~5000 visitors). Shared colour scale.



*Top:* the museum as-is.

*Bottom:* all eleven interventions, same colour scale.

The masterpiece bands (A9, A36) cool from red toward orange and the day spreads over more hours.

+31% welfare, revenue held.

Thank you.

Questions?

# Appendix

Backup slides for questions

## Backup: the three MDPs at a glance

| Setting                         | State         | Action                                | Reward           | $\gamma$    |
|---------------------------------|---------------|---------------------------------------|------------------|-------------|
| Toy (Q, Double-Q)               | 5-tuple table | stay / move                           | toy per-step     | 0.99        |
| As-is (PPO, DQN)                | 8 numbers     | stay / move on                        | walk, no booking | 0.995/0.997 |
| RAMA (MaskablePPO,<br>PPO, DQN) | 25 numbers    | lead / book / pace<br>masked for MPPO | booking per-step | 0.999       |

*The same five elements in every setting; what changes is the problem the agent is attached to, plus how the state is stored and the actions screened. PPO and DQN appear twice, once as a no-booking walk (the baselines) and once as booking agents; MaskablePPO only on RAMA.*

## Backup: the as-is state, eight numbers

The visitor's local view on the planned walk (natural units; the network sees each scaled to about  $[0, 1]$ ):

- 1 *Room-value drain*: minutes absorbed vs the room's worth ( $\sim 3$  min per importance point).
- 2 *Room value*: importance, 1 to 10.
- 3 *Density*: how full the room is, percent of capacity.
- 4 *On-target flag*: are you in the next planned must-see?
- 5 *Plan completed*: share of must-sees finished.
- 6 *Time of day*: minute of the 615-minute day.
- 7 *Egress slack*: spare minutes to reach an exit before closing.
- 8 *Distance to next room*: doorways to the next must-see.

*At the door A1*: (0, 0.22, 0.30, 0, 0, 0, 0.99, 0.10).



## Backup: the RAMA state, twenty-five numbers

A two-phase episode (a booking screen, then the walk), grouped:

- *Booking position* (3): phase flag, which of 4 decisions, lead chosen (days).
- *Booking screen* (4): slots free at lead 1/7/21/35 days (out of 8).
- *Time of day* (1): minute of the 720-minute day.
- *Windows held* (3): booked clock-times for Botticelli, Leonardo, Raphael.
- *Check-ins* (3): which masterpieces collected.
- *Access read* (4): is-masterpiece, unbooked, open-now, minutes-to-open.
- *Pace prior* (3): when it expects to reach each masterpiece (frozen).
- *Room tail* (4): drain, value, density, egress, as in the as-is state.

*Opening screen: (1, 0, 1; 0.12, 0.38, 0.62, 1; 0; ...); the booking screen is the crux.*

### Actions per setting

- Toy / as-is: Discrete(2), stay or move on.
- RAMA: Discrete(4). Booking phase: a lead  $\in \{1, 7, 21, 35\}$  or book/decline.  
Walk phase: stay or move on.

**Masking (MaskablePPO).** Set the logits of illegal moves to  $-\infty$  before the softmax, so they get exactly zero probability:

$$\pi(a \mid s) = \frac{e^{z_a} \mathbf{1}[a \in \mathcal{A}(s)]}{\sum_b e^{z_b} \mathbf{1}[b \in \mathcal{A}(s)]}.$$

## Backup: the reward functions

**Toy, per step.**  $\frac{\text{importance} \times \text{novelty}}{1 + \alpha_A \text{density}^2} - \text{step cost}$ , with  $\alpha_A = 6$ , step 0.5, exit bonus +15, no-exit penalty -50 (hand-tuned shaping; the relative sizes are what matter).

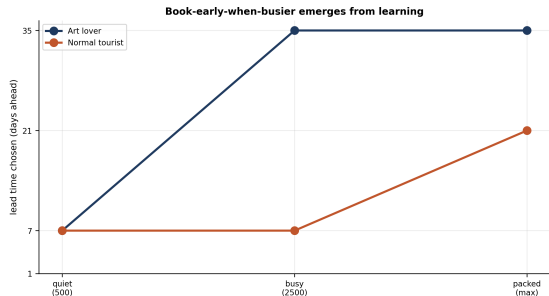
**Full museum, per minute.**  $\text{importance} \times \text{crowd discount} \times \text{novelty (satiation)}$ , with a gentle crowd  $\alpha = 0.5$ , plus closing-time pressure, a completion bonus and a small step cost.

**Booking terms (RAMA only).** an *immediate off-pace penalty* at commit time (a local proxy for the distant cost of a badly-timed slot) plus a no-show penalty for a masterpiece booked but missed.

## Two-phase episode

- Booking (no time passes): pick a *lead time* (1/7/21/35 days), then book each masterpiece.
- Windows committed near the expected arrival; then the minute-by-minute walk.

**Fill curve (busy day).** Slots free out of 8: 1 a day before, 3 a week ahead, 5 three weeks ahead, 8 a month ahead. Book late, miss the masterpiece.



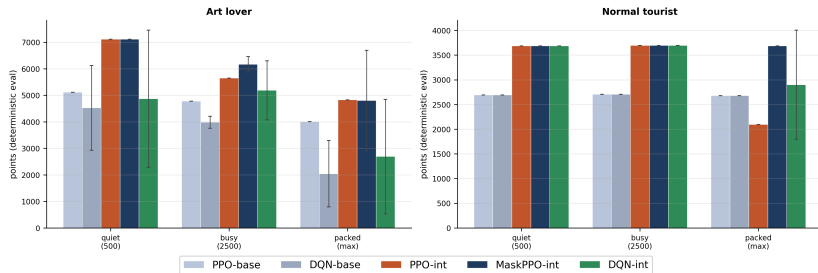
## Backup: full booking results, five seeds (mean $\pm$ std)

| Profile | Crowd | Baseline | Intervened      | Gain | Check-ins |
|---------|-------|----------|-----------------|------|-----------|
| Art     | 500   | 5112     | $7113 \pm 0$    | +39% | 3.0/3     |
| Art     | 2500  | 4780     | $6167 \pm 291$  | +29% | 3.0/3     |
| Art     | 5000  | 4010     | $4800 \pm 1891$ | +20% | 2.6/3     |
| Tourist | 500   | 2694     | $3686 \pm 0$    | +37% | 3.0/3     |
| Tourist | 2500  | 2708     | $3695 \pm 0$    | +36% | 3.0/3     |
| Tourist | 5000  | 2680     | $3685 \pm 0$    | +38% | 3.0/3     |

*Baseline = best no-booking walk (ppo\_base); intervened = MaskablePPO on RAMA. Five training seeds {1, 3, 7, 42, 202606}, each evaluated on six fixed common-random-number crowds. The art lover at max crowd is bimodal: four seeds reach  $\sim +41\%$ , one collapses, hence the wide band and the 2.6/3 check-ins.*

# Backup: which algorithm and where it breaks

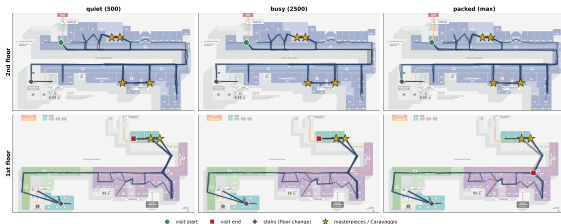
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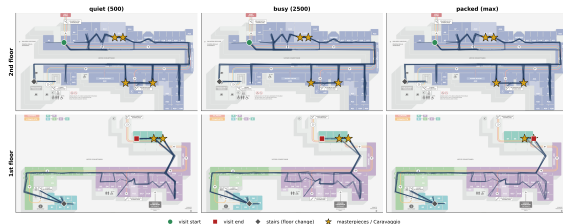
- *MaskablePPO*: best or tied in every cell, near-zero seed variance.
- *Unmasked PPO*: collapses at the packed tourist to 2093, below the 2680 no-booking baseline, while MaskablePPO holds 3685.
- *DQN*: crowd-fragile and high-variance, its art baseline falls 4527  $\rightarrow$  2044 as the crowd grows (a max over noisy values overestimates).

# Backup: the learned walk, art lover

Art lover walking routes (no interventions) – one continuous visit, 8 rollouts/panel; it descends floor 2 -> floor 1.



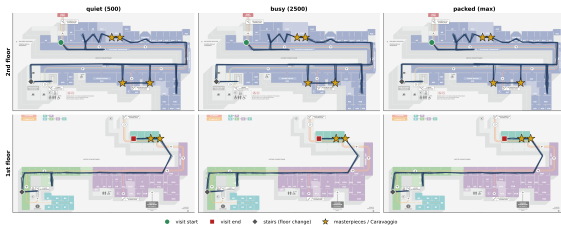
Art lover walking routes (RAMA + interventions) – one continuous visit, 8 rollouts/panel; it descends floor 2 -> floor 1.



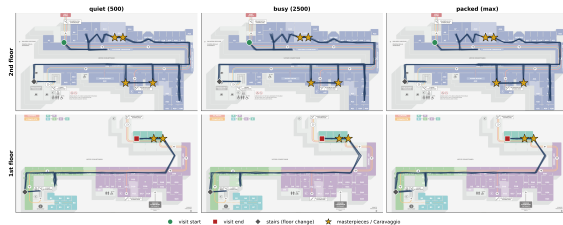
Left: as-is. Right: under RAMA. Eight rollouts per panel across the three crowd levels; the masterpieces (gold stars) are taken in their booked windows, then a descent to Caravaggio and out.

# Backup: the learned walk, tourist

Normal tourist walking routes (no interventions) -- one continuous visit, 8 rollouts/panel; it descends floor 2 -> floor 1.



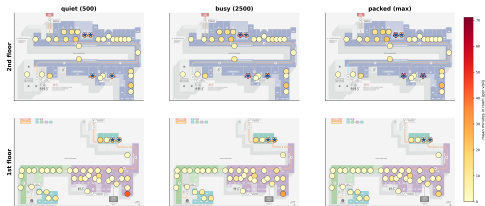
Normal tourist walking routes (RAMA + interventions) -- one continuous visit, 8 rollouts/panel; it descends floor 2 -> floor 1.



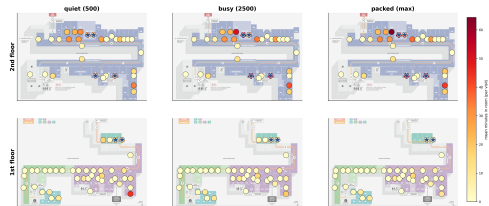


# Backup: per-profile crowd density

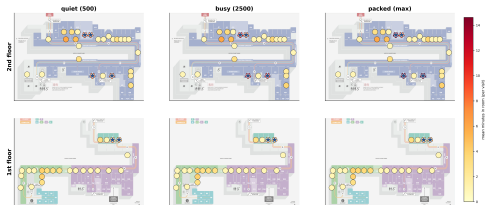
Art lover -- where time is spent (no interventions). Darker = more minutes. Blue stars = masterpieces/Caravaggio.



Art lover -- where time is spent (RAMA + interventions). Darker = more minutes. Blue stars = masterpieces/Caravaggio.



Normal tourist -- where time is spent (no interventions). Darker = more minutes. Blue stars = masterpieces/Caravaggio.

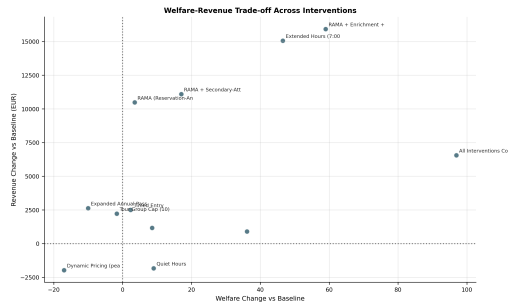


Normal tourist -- where time is spent (RAMA + interventions). Darker = more minutes. Blue stars = masterpieces/Caravaggio.

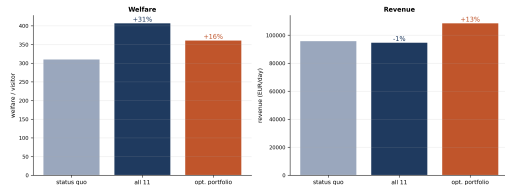


*Left column as-is, right column under RAMA; art lover on top, tourist below. The booked windows spread each profile's masterpiece visits across the day.*

# Backup: the interventions and the director's gate

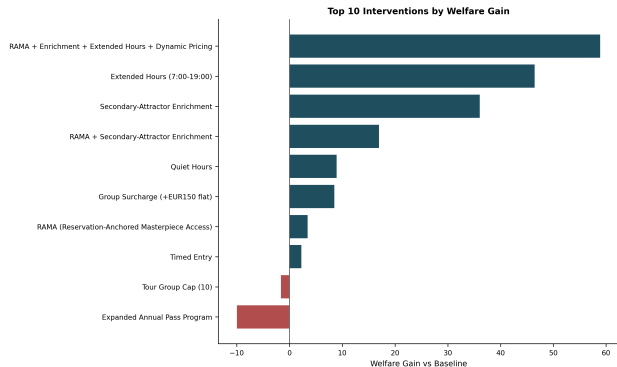


Museum-wide director gate (as-is crowd): all 11 lift welfare with revenue preserved; the optimized subset raises both



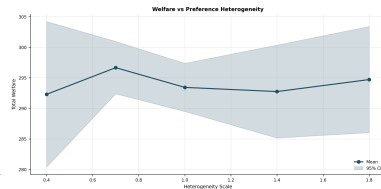
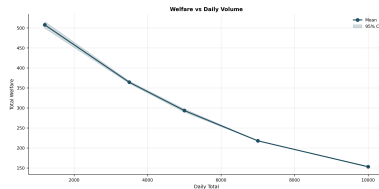
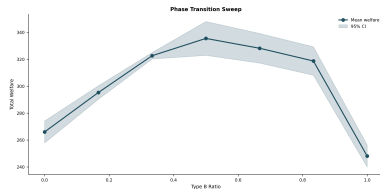
The redesign is adopted only if welfare rises and revenue does not fall (the director's gate). The chosen portfolio of eleven Pigovian measures lifts welfare **+31%** with revenue preserved.

## Backup: ranking the candidate interventions by welfare gain



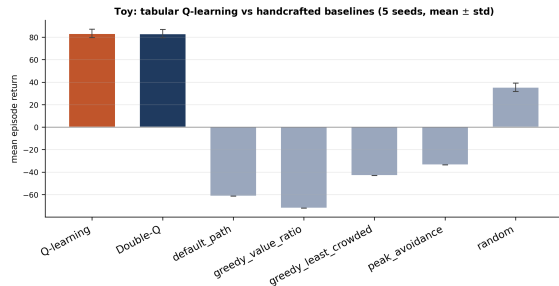
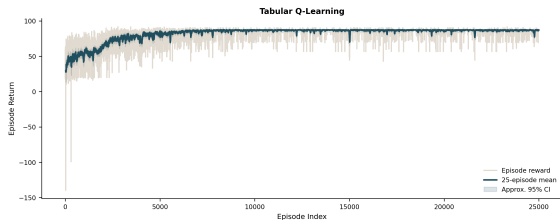
Each candidate measure scored on its own against the as-is baseline. Extended opening hours and secondary-attractor enrichment are the strongest single levers; the full combined bundle leads. The tour-group cap and the expanded annual pass lower welfare on their own; the adopted portfolio keeps the welfare-positive measures.

# Backup: robustness sweeps



We vary the Type-B (Instagram) share, the total daily volume and the crowd heterogeneity. The welfare gain from the redesign holds across the ranges we tried.

# Backup: tabular toy internals



Left: the learning curve, return climbing over episodes. Right: learning beats every hand-written rule. Across five seeds Q ( $83.3 \pm 3.8$ ) and Double-Q ( $83.1 \pm 3.7$ ) tie, since a deterministic toy has no overestimation bias for Double-Q to remove.

**Deep RL (Stable-Baselines3).** MlpPolicy [128, 128], Adam LR  $3 \times 10^{-4}$ , 2048 steps, batch 256, 10 epochs, GAE  $\lambda = 0.98$ , clip 0.2, entropy 0.01.

- As-is baselines (PPO): 150k steps,  $\gamma$  0.995/0.997.
- RAMA arms: 100k steps,  $\gamma = 0.999$ .

**Toy.** tabular Q / Double-Q,  $\varepsilon$ -greedy,  $\gamma = 0.99$ , 25k episodes.

### Seeds and evaluation

- 5 training seeds {1, 3, 7, 42, 202606}.
- Eval on 6 fixed crowds (900000 to 900005), deterministic, common random numbers.
- Crowd levels 500 / 2500 / 5000 visitors a day.

## Backup: limitations and where we would go next

**One visitor, not a crowd.** What we optimise is a single sophisticated visitor's best response to a *fixed* crowd. It is not an equilibrium: if every visitor booked this way the good slots would fill faster and the fill curve itself would move. Closing that loop, a mean-field game over the whole population, is the real next step.

**Fixes we did not run.** Our toy isolates the overestimation that makes DQN crowd-fragile, yet we never tried the obvious cure, Double-DQN, on the full task. Two experiments would sharpen the story: a fixed-heuristic booker (“always reserve a month out”) as a tougher baseline than no-booking, plus an ablation that switches the off-pace penalty off to confirm that booking early is learned rather than coaxed.

**What is baked in.** The connoisseur always books (a trait, not a trick to win); lead times are four discrete values; the fill curve is calibrated rather than fitted to real booking data; the policy is a plain feedforward net carrying appreciation-progress, with  $\gamma = 0.999$ . Welfare, finally, is the simulator's own visitor-utility, itself a modelling choice rather than a settled social objective.